

IMO 2026 — Problem 4

Solution by Meta Muse Spark 1.1

2026-07-21

Source: `results/muse-spark-1.1/problem-04/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ , then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Solution document

Status

solved ## Problem Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ , then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Approaches tried

- Modeled game as angle triples (p, q, r) , $p + q + r = 180$. Move: donor D , split $D = x + (D - x)$, children $(p, x, 180 - p - x)$ and $(q, D - x, p + x)$.
- Computed iterative winning closure for integer angles: W_0 contains θ , W_1 contains 2θ (or all if $\theta = 90$). Closure reaches all triangles iff $180/\theta$ integer.
- Proved sum closure for universally winning angles.

- For $180 = n\theta$, gave explicit two-phase strategy: force a multiple, then decrease multiple by splitting off θ .
- For 180 not multiple of θ , proved invariant I of triangles with no $k\theta$ angle; adversary can stay in I .

Current best

Mulan wins uniformly iff $\theta = 180/n$, $n \geq 2$ integer. If $\theta = 90^\circ$, one move suffices; if $\theta = 180/n$, $n \geq 3$, $n - 1$ moves suffice. Otherwise Shan-Yu chooses initial triangle in $I = \{\text{no angle } k\theta\}$ and preserves membership.

Full proof

Let a triangle be identified with its (unordered) angle triple (p, q, r) , $p, q, r > 0$, $p + q + r = 180$. A move: choose a donor $D \in \{p, q, r\}$, say $T = (p, q, D)$ with p, q the other two, choose $x \in (0, D)$ and produce

$$(1)C_1 = (p, x, q + D - x), \quad C_2 = (q, D - x, p + x) \quad (1)$$

since $180 - p - x = q + D - x$. Shan-Yu chooses which C_i survives. Put $W_0 = \{T : \theta \in T\}$, and

$$W_{k+1} = W_k \cup \{T \notin W_k : \exists \text{ donor } D, x \text{ with } C_1, C_2 \in W_k\}.$$

Mulan can force win in $\leq k$ moves uniformly iff $T \in W_k$. Put $L_k = W_k^c$.

1. First move

Let $T = (p, q, D) \notin W_0$. When $\exists x$ with $C_1, C_2 \in W_0$? $C_1 \in W_0 \iff x = \theta$ or $180 - p - x = \theta$; $C_2 \in W_0 \iff D - x = \theta$ or $p + x = \theta$. Four possibilities:

- $x = \theta$, $D - x = \theta \Rightarrow D = 2\theta$.
- $x = \theta$, $p + x = \theta \Rightarrow p = 0$ impossible.
- $180 - p - x = \theta$, $D - x = \theta \Rightarrow D = 180 - p - \theta$ gives $q + D - x = \theta$, with $D - x = \theta$ gives $q = 0$ impossible.
- $180 - p - x = \theta$, $p + x = \theta \Rightarrow 180 = 2\theta$.

Hence for $\theta \neq 90^\circ$, $W_1 \setminus W_0$ is exactly triangles containing angle 2θ (if $2\theta < 180$), otherwise empty.

For $\theta = 90^\circ$ every triangle is in W_1 : let T have angles $p \leq q \leq D$ (or any order). At least two are acute; pick $p, q < 90$. $p > 0$ and $q < 90$ implies $D + p = 180 - q > 90$, so $r := 90 - p$ satisfies $0 < r < D$. Splitting D as r and $D - r$ gives $C_1 = (p, r, 180 - p - r) = (p, 90 - p, 90)$, $C_2 = (q, D - r, p + 90 - p) = (q, D - 90 + p, 90)$. Both contain 90° . Thus Mulan wins in one move. This is the case $n = 2$.

Assume henceforth $\theta \neq 90^\circ$.

2. Multiples become universally winning

Call ϕ universally winning at level k if every triangle containing ϕ lies in W_k . θ is universally winning at level 0. If s, s' are universally winning at level k and $s + s' < 180$, then $s + s'$ is

universally winning at level $k + 1$: any $T = (p, q, s + s')$ with $x = s$ gives C_1 containing s , C_2 containing s' , both in W_k , so $T \in W_{k+1}$.

Inductively, if $k\theta < 180$ then $k\theta$ is universally winning and

$$(2)\{\text{triangles containing } k\theta\} \subset W_{k-1}, \quad (2)$$

with W_0 for $k = 1$. Indeed $k\theta = (k - 1)\theta + \theta$.

Explicit decrease: if T contains $k\theta \geq 2\theta$, cutting $k\theta$ as $\theta + (k - 1)\theta$ yields C_1 with θ , C_2 with $(k - 1)\theta$. Whichever Shan-Yu keeps, either Mulan wins immediately or the new triangle contains $(k - 1)\theta$.

Let $m = \lfloor 180/\theta \rfloor$, $\mathcal{M} = \{k\theta : 1 \leq k \leq m, k\theta < 180\}$ with $|\mathcal{M}| = m$ unless 180 multiple. After $m - 1$ steps W_{m-1} contains all triangles whose angle is in $\mathcal{M}$.

3. Divisor case $\theta = 180/n$ Mulan wins

Suppose $180 = n\theta$, $n \geq 2$ integer; $n = 2$ done. Let $n \geq 3$, so $\theta \leq 60^\circ$. Then $\mathcal{M} = \{\theta, 2\theta, \dots, (n-1)\theta\}$, $(n-1)\theta = 180 - \theta$, and $180 - k\theta = (n - k)\theta \in \mathcal{M}$. Any triangle containing a multiple is already in W_{n-2} by (2).

Let T contain no angle from $\mathcal{M}$. Every triangle has an angle $\leq 60^\circ$; choose p minimal, so $p \leq 60^\circ$ and $p \notin \mathcal{M}$. Write $p = k\theta + \alpha$ with $0 < \alpha < \theta$, $k \geq 0$; then $r := (k + 1)\theta - p = \theta - \alpha \in (0, \theta)$ and $p + r = (k + 1)\theta \in \mathcal{M}$. Since $n \geq 3$, $p + r \leq 60^\circ + \theta \leq 120^\circ < 180$.

Let D be a maximal angle of T , $D \geq 60^\circ \geq r$, indeed $r < \theta \leq 60^\circ \leq D$; also $D \notin \mathcal{M}$, so $D > r$ unless $p + r$ trivial, but $0 < r < D$ holds after choosing p minimal, so $x = r$ is feasible. Cutting D as r and $D - r$:

$$C_1 = (p, r, 180 - p - r), \quad C_2 = (q, D - r, p + r)$$

where q is the third angle. $p + r \in \mathcal{M}$ hence C_2 contains a multiple. $180 - p - r = 180 - (k + 1)\theta = (n - k - 1)\theta \in \mathcal{M}$ (as $180 = n\theta$), hence C_1 contains a multiple. Thus both children are in W_{n-2} .

Therefore $T \in W_{n-1}$. Mulan's uniform strategy: - if current triangle contains $k\theta$, $k \geq 2$, cut $k\theta$ into $\theta + (k - 1)\theta$; Shan-Yu loses or k drops by one. - if it contains no multiple, pick p minimal, D maximal, cut D with $r = \lceil p/\theta \rceil \theta - p$; both children contain multiples, so after this move we are in previous case with $k \leq n - 1$.

Hence after at most $n - 1$ moves θ appears, whatever Shan-Yu does. Thus for $\theta = 180/n$ Mulan guarantees finite win.

4. Non-divisor case Shan-Yu avoids forever

Assume $\delta := 180 \bmod \theta \in (0, \theta)$, i.e. $180 = q_0\theta + \delta$, $0 < \delta < \theta$, so 180 is not a multiple of θ . Let $\mathcal{M}$ as above, $m = \lfloor 180/\theta \rfloor$, and

$$I = \{T : \text{no angle of } T \text{ belongs to } \mathcal{M}\}.$$

Since $\mathcal{M}$ finite, $I \neq \emptyset$: choose $a \notin \mathcal{M}$ small, choose $b \notin \mathcal{M}$ with $180 - a - b \notin \mathcal{M}$.

Note $\theta \in \mathcal{M}$, so $I \subset L_0$.

Lemma: If $T = (p, q, D) \in I$ then for any choice of donor and any $x \in (0, D)$ at least one of the two children in (1) lies in I .

Proof: Suppose $T \in I$, i.e. $p, q, D \notin \mathcal{M}$, and both children $\notin I$. Then each contains an angle from $\mathcal{M}$. Since $p, q \notin \mathcal{M}$, this forces

- for C_1 : $x = k_1\theta$ or $180 - p - x = k_1\theta$,
- for C_2 : $D - x = k_2\theta$ or $p + x = k_3\theta$.

Four combos:

1. $x = k_1\theta, D - x = k_2\theta \Rightarrow D = (k_1 + k_2)\theta \in \mathcal{M}$, contradiction.
2. $x = k_1\theta, p + x = k_3\theta \Rightarrow p = (k_3 - k_1)\theta$, multiple (or zero if $k_3 = k_1$, impossible as $p > 0$).
3. $180 - p - x = k_1\theta, D - x = k_2\theta \Rightarrow 180 - p - D = (k_1 - k_2)\theta$, i.e. $q = (k_1 - k_2)\theta \in \mathcal{M}$.
4. $180 - p - x = k_1\theta, p + x = k_3\theta \Rightarrow (k_1 + k_3)\theta = 180$, so 180 multiple of θ , contradicting $\delta \neq 0$.

All impossible, so at least one child stays in I .

Consequence: Shan-Yu can forever stay in I . Starting from $T_0 \in I$, whatever donor D and split x Mulan announces, Shan-Yu chooses the child belonging to I (exists by lemma). The game never hits an angle in $\mathcal{M}$, in particular never hits θ . Hence Mulan has no finite winning strategy; indeed she never wins.

This includes $\theta > 90^\circ$, $\theta \neq 90^\circ$, where $\mathcal{M} = \{\theta\}$ and the same case analysis shows $W_1 = W_0$.

5. Answer

Mulan can force victory in boundedly many steps iff $180/\theta$ is an integer. I.e.

$$\theta = \frac{180^\circ}{n} \text{ for some integer } n \geq 2.$$

For $n = 2$ ($\theta = 90^\circ$) one cut suffices; for $n \geq 3$ at most $n - 1$ cuts suffice by the strategy above (force a multiple, then descend). If θ is not of this form, Shan-Yu starts with a triangle having no angle $k\theta$ and can keep this property forever, so Mulan can never force θ .